

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Smart Lab Equipment & Slot Reservation Portal (SLESR)

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Project Key	SLESR
Epics	5
User Stories	14
Total Story Points	49
Sprints Completed	2

1. Epics & Backlog (Story Points, Priority, Status)

The backlog was organized into 5 Epics, each tracing directly back to a Functional Requirement from the Lab 1 requirements specification. Each Epic's child work items (User Stories) below show their assigned Story Points, Priority, and current Status.

Epic 1: Equipment Discovery & Calibration Assurance

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Description

Enable students to search the equipment catalogue and see real-time calibration status, so that only valid, calibrated equipment is ever offered for booking. (Realises FR-002)

Child work items

100% Done

Work	Pri...	Stor...	As...	Status
☒ SLESR-6 Story 1.1: Catalogue Search	^ Hi	3	U	Done ▾
☒ SLESR-7 Story 1.2: Calibration Visibility	^ Hi	2	U	Done ▾
☒ SLESR-8 Story 1.3: Hide Invalid Equipment	^ Hi	2	U	Done ▾

Epic 1: Equipment Discovery & Calibration Assurance (FR-002) — 100% Done

Epic 2: Slot Reservation

Description

Allow authenticated students to reserve a slot on calibrated equipment, atomically and without double-booking. (Realises FR-001, NFR-001)

Child work items

100% Done

Work	Pri...	Stor...	As...	Status
☒ SLESR-9 Story 2.1: Reserve a Slot	Hi	8	U:	Done ▾
☒ SLESR-10 Story 2.2: Reservation Token	Hi	3	U:	Done ▾
☒ SLESR-11 Story 2.3: Clean Conflict Handling	Hi	5	U:	Done ▾

Epic 2: Slot Reservation (FR-001, NFR-001) — 100% Done

Epic 3: Catalogue & Maintenance Management

Description

Give lab technicians control over the equipment catalogue and calibration/maintenance events, keeping the schedule in sync with real asset state. (Realises FR-003)

Child work items

0% Done

Work	Pri...	Stor...	As...	Status
SLESR-12 Story 3.1: Manage Equipment Records	Hi	5	Ui	To Do
SLESR-13 Story 3.2: Log Calibration Events	Hi	5	Ui	To Do
SLESR-14 Story 3.3: Cancellation Notifications	Hi	3	Ui	To Do

Epic 3: Catalogue & Maintenance Management (FR-003) — 0% Done

Epic 4: Checkout, Return & Penalty Enforcement

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Description

Track equipment checkout/check-in against reservations and apply consistent late-return penalties. (Realises FR-004)

Child work items

0% Done

Work	Pri...	Stor...	As...	Status
SLESR-15 Story 4.1: Scan Checkout and Check-in	Hi	5	Ui	To Do
SLESR-16 Story 4.2: Auto-Apply Late Return Penalty	Hi	5	Ui	To Do

Epic 4: Checkout, Return & Penalty Enforcement (FR-004) — 0% Done

Epic 5: Cancellation & Waitlist



Description

Let students cancel reservations and join a fair waitlist queue for fully booked equipment.
(Realises FR-005)

Child work items

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Work		Pri...	Stor...	As...	Status
☒ SLESR-17	Story 5.1: Cancel Reservation	 M	2	 UI	To Do ▾
☒ SLESR-18	Story 5.2: Join Waitlist	 M	5	 UI	To Do ▾
☒ SLESR-19	Story 5.3: Waitlist Claim Window	 M	3	 UI	To Do ▾

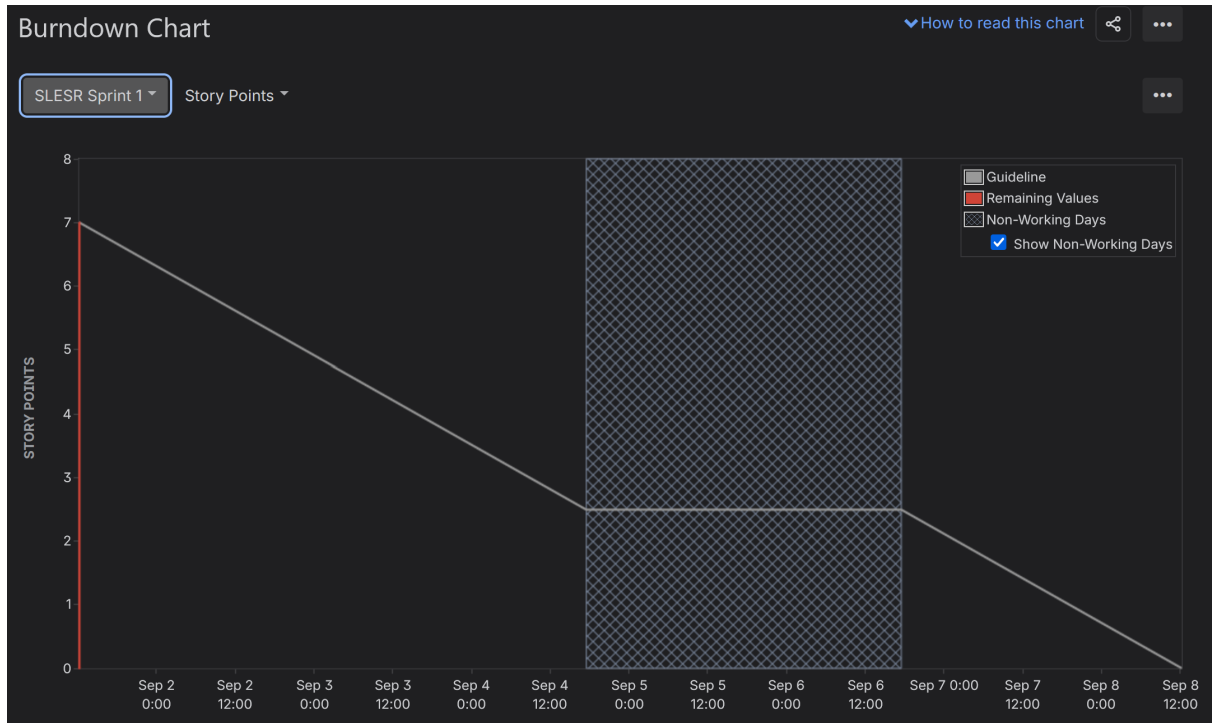
0% Done

Epic 5: Cancellation & Waitlist (FR-005) — 0% Done

2. Sprint Execution & Burndown Charts

Two sprints were run against the backlog. Sprint 1 covered Epic 1 (Equipment Discovery & Calibration Assurance) and Sprint 2 covered Epic 2 (Slot Reservation). Both sprints were started, worked through To Do → In Progress → Done, and completed.

2.1 Sprint 1 Burndown Chart



Sprint 1 (SLESR-6, 7, 8 — 7 story points total) shows a clean declining burndown from 7 points down to 0, tracking closely against the guideline.

2.2 Sprint 2 Burndown Chart



Sprint 2 (SLESR-9, 10, 11 — 16 story points total) shows an instantaneous drop from 16 points to 0 right at sprint start, since all three stories were moved to Done within minutes of starting the sprint during the simulation.

2.3 Sprint Change Logs (Story Point Burndown Detail)

Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
01/Sep/26 12:13 PM	SLESR-6	Sprint start		3		
	SLESR-7			2		
	SLESR-8			2		
	Total					7
01/Sep/26 12:14 PM	SLESR-6	Burndown	Issue completed		3	4
	SLESR-7	Burndown	Issue completed		2	2
	SLESR-8	Burndown	Issue completed		2	0
	SLESR-6	Sprint ended by Siddanth Anil				0
	SLESR-7					0
	SLESR-8					0
	Total					0

Sprint 1 change log — scope added (3+2+2=7 pts) then burned down to 0.

Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
01/Sep/26 12:14 PM	Total	Sprint start				0
	SLESR-9	Scope change	Issue added to sprint	8		8
	SLESR-10	Scope change	Issue added to sprint	3		11
	SLESR-11	Scope change	Issue added to sprint	5		16
01/Sep/26 12:16 PM	SLESR-9	Burndown	Issue completed		8	8
	SLESR-10	Burndown	Issue completed		3	5
	SLESR-11	Burndown	Issue completed		5	0
01/Sep/26 12:17 PM	SLESR-10	Sprint ended by Siddanth Anil				0
	SLESR-11					0
	SLESR-9					0
	Total					0

Sprint 2 change log — scope added (8+3+5=16 pts) then burned down to 0.

3. Reflection Questions

1. Did your estimations reflect the actual effort?

Story points were assigned using the Fibonacci scale based on the complexity implied by each story's acceptance criteria — for example, Story 2.1 (Reserve a Slot) was estimated at 8 points due to the concurrency and validation logic (2-hour cap, 7-day horizon, calibration check), while simpler stories like Story 1.2 (Calibration Visibility) were estimated at 2 points. Since this was a simulated sprint rather than actual development work, the estimates could not be validated against real implementation time — both sprints completed all planned points almost immediately after starting, so the estimates were never actually tested against real effort. In a live project, the true test would be whether 8-point stories consistently took roughly proportionally longer than 2-point stories.

2. Was your backlog well-prioritized?

Yes. Priorities were derived directly from the Lab 1 requirements table: Epics 1–4 (covering FR-001 through FR-004, all rated High priority in the original requirements specification) were marked High priority in Jira, while Epic 5 (Cancellation & Waitlist, covering FR-005, rated Medium priority) was marked Medium. This kept the ordering consistent with the original business-value ranking rather than introducing a new, arbitrary prioritization scheme at the backlog stage.

3. How did your simulated sprint align with your plan?

Both sprints completed 100% of their planned scope — Sprint 1 finished all 3 stories from Epic 1 (7 points) and Sprint 2 finished all 3 stories from Epic 2 (16 points), with no scope changes mid-sprint beyond the initial addition of stories at sprint start. However, the alignment was artificial in one respect: since this was a simulation rather than a real week of development, all stories were moved from To Do to Done within minutes rather than being spread across the full sprint duration. This means the sprints technically met their planned scope, but did not exercise the day-to-day pacing a real sprint would.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts reveal a pattern the handout specifically calls out: work was completed far ahead of the guideline line in both sprints. Sprint 1's chart shows a smoother decline since a few minutes elapsed between completing individual stories, but Sprint 2 shows an instant vertical drop from 16 points to 0 right at the start. Per the handout's guidance, a team that consistently finishes work early is a sign they aren't committing to enough work during sprint planning — in a real team setting, this would suggest that either the sprint duration was set too conservatively for the actual scope, or that more stories per sprint should have been planned in to make full use of the available capacity. It also reinforces that burndown data is only meaningful when sprint activity is genuinely distributed over time, which a simulated same-day sprint cannot fully demonstrate.